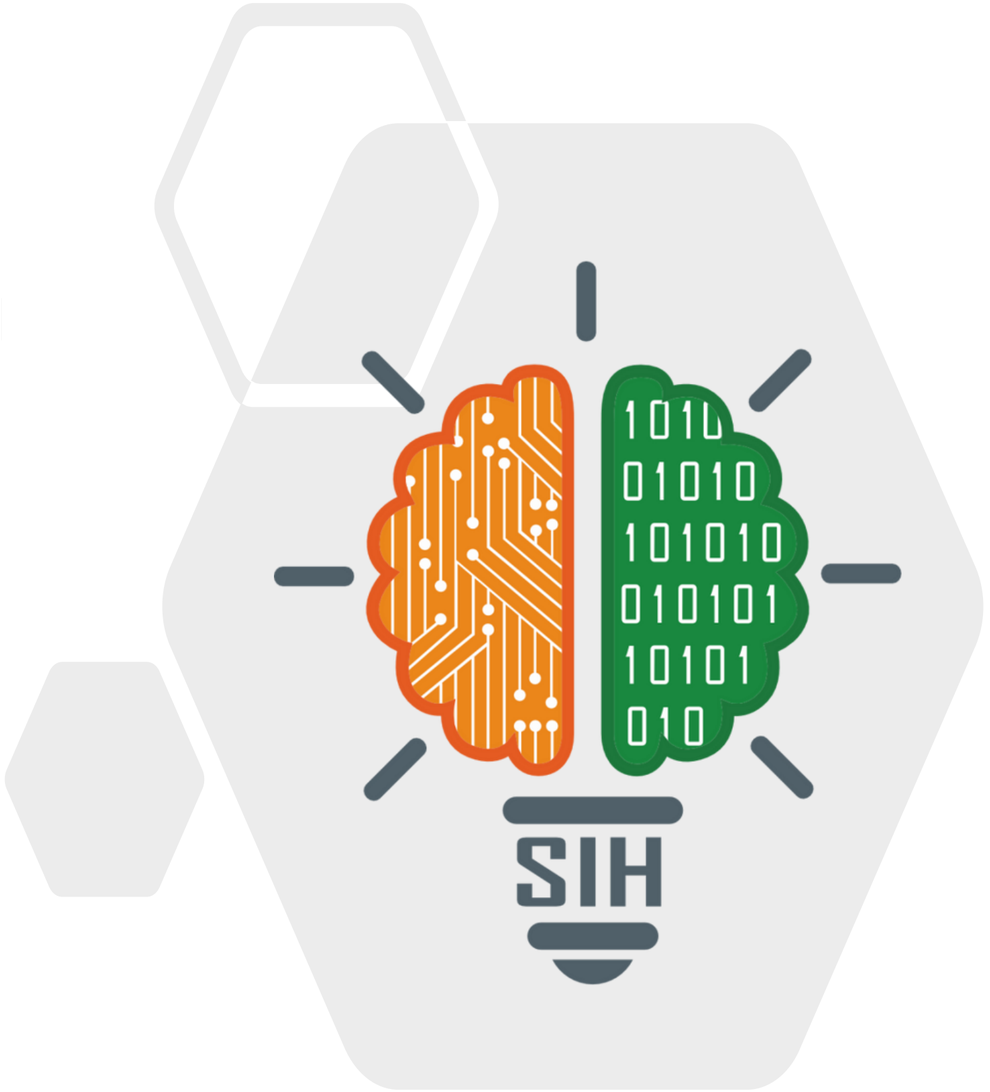
SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID -** 25203
- **Problem Statement Title -** AI-Powered Multilingual content localization engine for skill courses
- **Theme -** Miscellaneous
- **PS Category -** Software
- **Team ID –** 112680
- **Team Name -** BotSlam



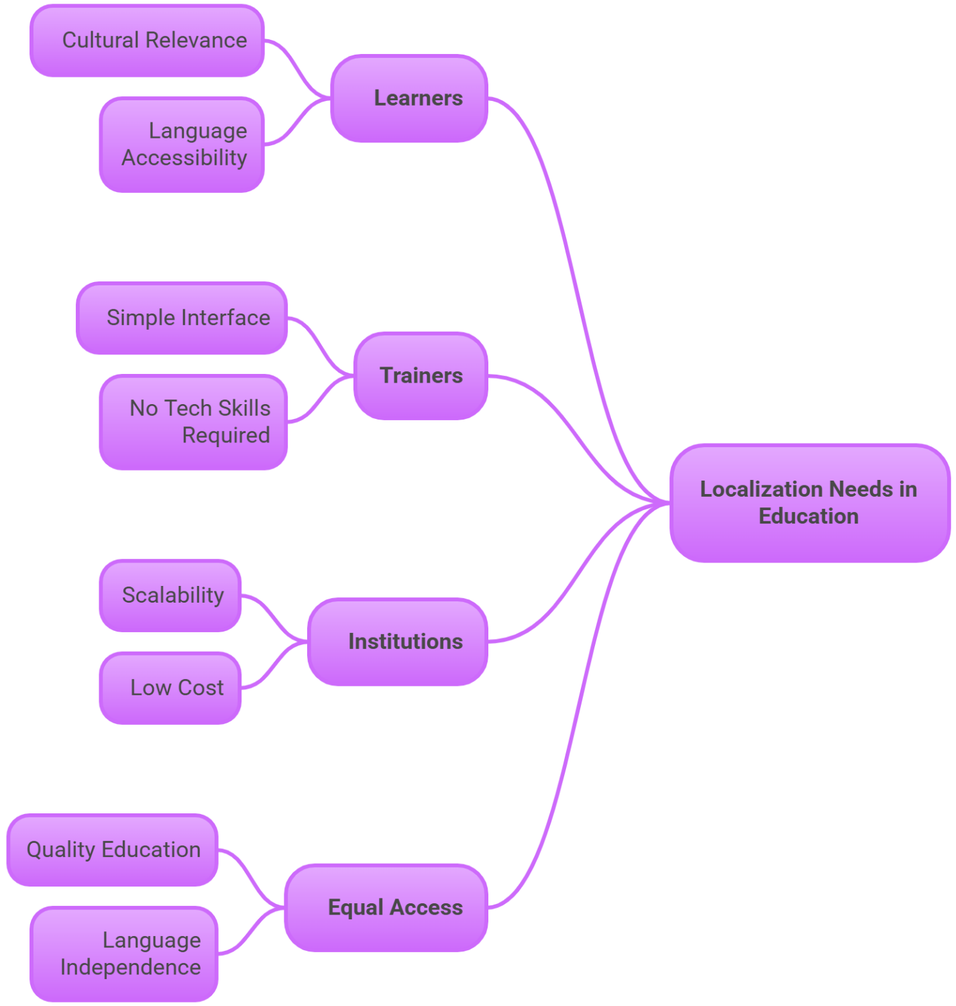
Proposed Solution

- We propose an AI-driven localization engine that translates and culturally adapts vocational training content into local Indian languages.
- It supports text, audio, and video content, offers speech-to-text and text-to-speech features, and integrates easily with Skill India Digital and LMS platforms.

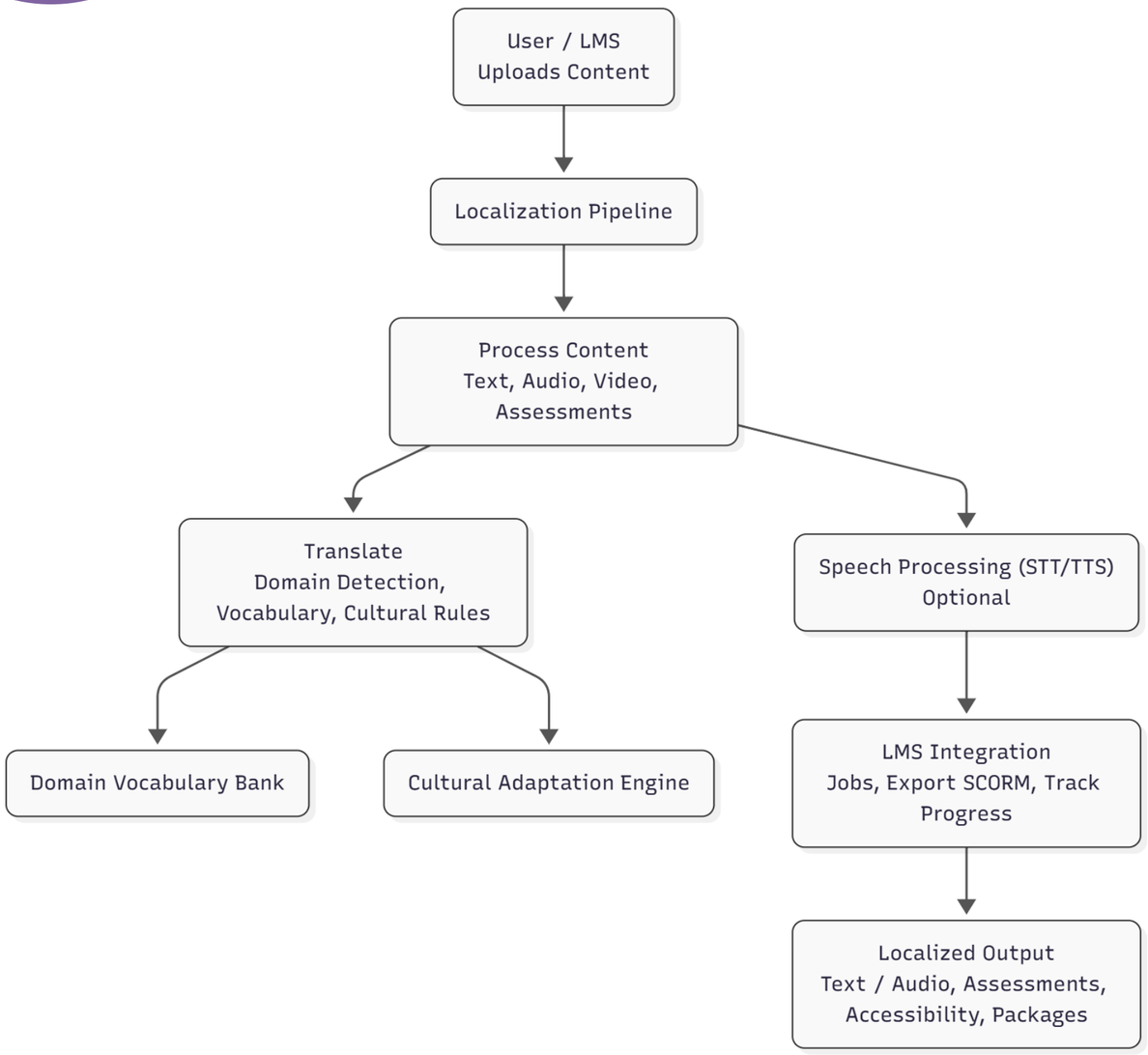
How It Solves the Problem

- Converts existing English training modules into multiple Indian languages.
- Uses domain-specific vocabulary for sectors like manufacturing, healthcare, IT, and agriculture.
- Adapts lessons to regional culture and examples, improving relatability.
- Saves time and cost compared to manual translation.
- Increases participation and understanding among rural and non-English learners.

Localization need in Education



TECHNICAL APPROACH



TECHSTACK

python™

JavaScript

Streamlit

FastAPI

GTTS

Whisper

GitHub

BHASHINI

FFmpeg

FEASIBILITY AND VIABILITY

Technical Feasibility

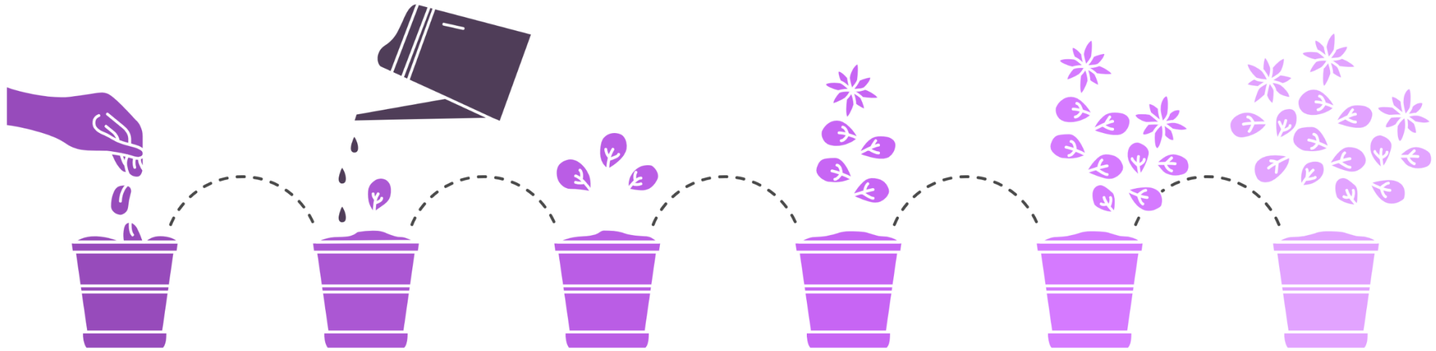
Feature	Description
NLP Models	Transformer-based models for accurate translation
Content Types	Text, audio, and video content processing
Terminology	Domain-specific glossaries for terminology accuracy
Accessibility	Speech-to-text and text-to-speech layers
Infrastructure	Cloud-native (AWS / Azure) for scalability

Viability

•**Scalable & Cost-Effective:**
Uses cloud-based deployment and open-source NLP models.
•**Government & Industry Support**:
Aligned with NCVET and MSDE's Digital India mission strong potential for policy backing and public-private partnerships.
•**Sustainable Model:**
generate revenue via API integrations with LMS platforms.

Challenge and risk

•Language Complexity: Regional dialects and mixed-code usage may affect accuracy.
•Contextual Consistency: AI may struggle with industry-specific phrasing.
•Data Availability: Limited localized training datasets for certain skill sectors.
•Quality Assurance: Continuous human validation required for early phases.
•Integration Barriers: Aligning APIs with existing government digital ecosystems.



English-Centric Skilling
80% of content in English, inaccessible to most

Cloud & NLP Integration
Affordable translation via cloud and open-source NLP

Trusted Frameworks
Leveraging proven translation platforms for reliability

Nationwide Implementation
Elastic architecture ensures no downtime

Government Alignment
Supporting inclusive skilling initiatives

Inclusive Skilling
Accessible skill content for non-English speakers

IMPACT AND BENEFITS

Benefits

Deployment Speed

Accelerates content rollout by automating translation processes.

Quality Learning

Maintains high standards through accurate and contextual AI translations.

Inclusivity

Provides tools for learners with disabilities to access education.



Accessibility

Ensures skilling content is available in multiple languages for all learners.

Cost Efficiency

Reduces expenses by automating manual translation and production.

Impact



Educational Impact

Bridges language barriers, expanding learner access.

Social Impact

Empowers rural learners, promoting equity and inclusion.

Economic Impact

Increases job readiness and workforce participation.

Technological Impact

Drives innovation in AI and education technology.

Governance & Policy Impact

Aligns with national skill development missions.

RESEARCH AND REFERENCES

•**National Language Translation Mission (Bhashini)** MeitY, Government of India (2023).

•**IndicNLP Suite: Natural Language Processing for Indian Languages**– AI4Bharat (2022)
Open-source NLP framework supporting multiple Indian languages for translation and text analysis.

•**AI in Vocatinal Training and Education UNESCO Report (2023**)
Highlights the role of AI in improving accessibility and inclusivity in skill-based education.

•**Digital Skilling Ecosystem: NCVET Annual Report– National Council for Vocational Education and Training (2023**)
Framework for integrating technology and multilingual accessibility in vocational programs.

•**Google Cloud Translation & Microsoft Translator Documentation (2024**)
Technical references for API integration and scalable AI-based translation pipelines.

•**AI for India 2.0: Skill Development Vision**– Ministry of Skill Development and Entrepreneurship (MSDE, 2024)
Policy direction on using AI tools to enhance India's digital skilling mission

